

The Maximum Likelihood Origin of Objective Functions in Supervised Learning

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November 18, 2025

A Note on Mathematical Notation

We distinguish between a *function* and its *value*. A function is a mapping, which we will define with its type signature and, where possible, its rule. For a function $f : \mathcal{X} \rightarrow \mathcal{Y}$, the expression $f(x)$ denotes the *value* of f evaluated at $x \in \mathcal{X}$.

1 Basic Definitions

Definition 1.1 (σ -algebra). *Let Ω be a non-empty set. A collection $\mathcal{F} \subseteq 2^\Omega$ is a σ -algebra if:*

1. $\Omega \in \mathcal{F}$
2. $A \in \mathcal{F} \implies A^c \in \mathcal{F}$
3. $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{F} \implies \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$

Definition 1.2 (Measurable Space). *A pair $(\Omega, \mathcal{F})$ where Ω is a set and $\mathcal{F}$ is a σ -algebra on Ω .*

Definition 1.3 (Probability Measure). *A function $P : \mathcal{F} \rightarrow [0, 1]$ is a probability measure if:*

1. $P(\Omega) = 1$
2. For disjoint $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{F}$: $P(\bigcup_{n=1}^{\infty} A_n) = \sum_{n=1}^{\infty} P(A_n)$

Definition 1.4 (Probability Space). *A triple $(\Omega, \mathcal{F}, P)$ where $(\Omega, \mathcal{F})$ is a measurable space and P is a probability measure on $\mathcal{F}$.*

Definition 1.5 (Measurable Function). *Given measurable spaces $(E, \mathcal{E})$ and $(F, \mathcal{F})$, a function $f : E \rightarrow F$ is measurable if:*

$$f^{-1}(B) \in \mathcal{E} \text{ for all } B \in \mathcal{F}$$

Definition 1.6 (Random Variable). *A measurable function $X : (\Omega, \mathcal{F}) \rightarrow (\mathcal{X}, \mathcal{E})$ from a probability space to a measurable space $(\mathcal{X}, \mathcal{E})$.*

Definition 1.7 (Pushforward Measure). *Given a measurable function $X : (\Omega, \mathcal{F}) \rightarrow (\mathcal{X}, \mathcal{E})$ and a measure P on $(\Omega, \mathcal{F})$, the pushforward measure P_X on $(\mathcal{X}, \mathcal{E})$ is:*

$$P_X(B) = P(X^{-1}(B)) = P(\{\omega \in \Omega : X(\omega) \in B\})$$

for all $B \in \mathcal{E}$.

Definition 1.8 (Distribution). *The pushforward measure P_X is called the distribution of the random variable X .*

Definition 1.9 (Density). Under assumed assumptions in ML, (if probability measure P_X is absolutely continuous with respect to σ -finite measure μ on $\mathcal{X}$ (written $P_X \ll \mu$), then (by the Radon-Nikodym theorem,) there exists a measurable function $p : \mathcal{X} \rightarrow [0, \infty)$ unique up to μ -null sets such that:

$$P_X(A) = \int_A p(x) d\mu(x)$$

for all measurable sets $A \subseteq \mathcal{X}$. We write $p = \frac{dP_X}{d\mu}$ and call it the density of X with respect to μ .

Remark 1.1 (ML Perspective). Densities enable gradient-based learning: the probability measure becomes a differentiable function $p(x)$, allowing optimization of log-likelihoods $\sum_i \log p(x_i)$ via backpropagation.

Example 1.1 (Bernoulli as Pushforward). Let $(\Omega, \mathcal{F}, P) = ([0, 1], \mathcal{B}([0, 1]), \text{Uniform}[0, 1])$ where $\mathcal{B}([0, 1])$ denotes the Borel σ -algebra. Define $X : [0, 1] \rightarrow \{0, 1\}$ by:

$$X(\omega) = \begin{cases} 1 & \text{if } \omega \leq p \\ 0 & \text{if } \omega > p \end{cases}$$

Then the pushforward measure is:

$$P_X(\{1\}) = P(\{\omega : X(\omega) = 1\}) = P([0, p]) = p$$

$$P_X(\{0\}) = P(\{\omega : X(\omega) = 0\}) = P((p, 1]) = 1 - p$$

Thus $P_X = \text{Bernoulli}(p)$ on $(\{0, 1\}, 2^{\{0, 1\}})$.

Remark 1.2 (The ML Perspective on Random Variables). In machine learning, we typically work with random variables by specifying only their distributions or observed values, without explicitly defining the underlying probability space $(\Omega, \mathcal{F}, P)$. This is justified because:

1. **Observational equivalence:** Two random variables with the same distribution are indistinguishable from observed data.
2. **Kolmogorov's theorem:** Given any probability measure μ on a measurable space, there exists a probability space and a random variable with distribution μ .
3. **Practical focus:** In ML, we care about the values we observe and the distribution that generated them, not the abstract sample space Ω .

A random variable is formally a function, but its distribution fully characterizes its probabilistic behavior. A *realization* is a single value we observe. We are given realizations to infer properties of the distribution.

2 Joint and Conditional Distributions

Example 2.1 (Joint Distribution: Two Coin Flips). Consider two independent fair coin flips with outcomes $X, Y \in \{H, T\}$. The joint distribution $P_{X,Y}$ is the probability measure on $\{H, T\} \times \{H, T\}$ that assigns:

$$P_{X,Y}(\{(H, H)\}) = 1/4, \quad P_{X,Y}(\{(H, T)\}) = 1/4, \quad P_{X,Y}(\{(T, H)\}) = 1/4, \quad P_{X,Y}(\{(T, T)\}) = 1/4$$

With respect to counting measure, the joint density is simply the probability mass function: $p_{XY}(x, y) = 1/4$ for all $(x, y) \in \{H, T\} \times \{H, T\}$. This satisfies $P_{X,Y}(\{(x, y)\}) = p_{XY}(x, y)$.

Definition 2.1 (Joint Distribution and Joint Density). For a pair of random variables $(X, Y) : \Omega \rightarrow \mathcal{X} \times \mathcal{Y}$, the joint distribution is the pushforward measure $P_{X,Y}$ on $\mathcal{X} \times \mathcal{Y}$.

If $P_{X,Y} \ll \mu \otimes \nu$ for product measure $\mu \otimes \nu$, the joint density $p_{XY} : \mathcal{X} \times \mathcal{Y} \rightarrow [0, \infty)$ satisfies:

$$P_{X,Y}(A \times B) = \int_A \int_B p_{XY}(x, y) d\nu(y) d\mu(x)$$

for all measurable $A \subseteq \mathcal{X}$, $B \subseteq \mathcal{Y}$.

Remark 2.1 (ML Perspective). While $p_{XY}(x, y)$ encodes the complete feature-label relationship, supervised learning models only the conditional factor $p_{Y|X}$ and treats the marginal p_X as unmodeled.

Definition 2.2 (Marginal Density). The marginal density of X is the function $p_X : \mathcal{X} \rightarrow [0, \infty)$ defined by:

$$p_X(x) = \int_{\mathcal{Y}} p_{XY}(x, y) d\nu(y)$$

This is the density of the marginal distribution P_X with respect to μ .

Remark 2.2 (ML Perspective). In supervised learning, we never estimate p_X —the feature distribution is treated as an arbitrary, unknown sampling process. We only model how Y depends on X .

Definition 2.3 (Conditional Density). For x such that $p_X(x) > 0$, the conditional density of Y given $X = x$ is the function $p_{Y|X}(\cdot|x) : \mathcal{Y} \rightarrow [0, \infty)$ defined by:

$$p_{Y|X}(y|x) = \frac{p_{XY}(x, y)}{p_X(x)}$$

Remark 2.3 (ML Perspective). This is what supervised learning models. Neural networks parameterize $p_\theta(y|x)$ by outputting distribution parameters: logits for categorical distributions, mean/variance for Gaussians, etc.

Proposition 2.1 (Chain Rule for Densities). The joint, marginal, and conditional densities satisfy:

$$p_{XY}(x, y) = p_{Y|X}(y|x) \cdot p_X(x)$$

Proof. Direct consequence of the definition: $p_{Y|X}(y|x) = \frac{p_{XY}(x, y)}{p_X(x)}$ implies $p_{XY}(x, y) = p_{Y|X}(y|x) \cdot p_X(x)$. □

Remark 2.4 (ML Perspective). This factorization justifies conditional-only modeling: we can estimate parameters θ in $p_\theta(y|x)$ from data without ever modeling $p_X(x)$.

3 Statistical Models

Definition 3.1 (Statistical Model). A triple $(\mathcal{X}, \mathcal{B}, \mathcal{P})$ where:

- $(\mathcal{X}, \mathcal{B})$ is a measurable space (the sample space)
- $\mathcal{P}$ is a collection of probability measures on $(\mathcal{X}, \mathcal{B})$

Definition 3.2 (Parametric Model). A statistical model where $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ with $\Theta \subseteq \mathbb{R}^d$ called the parameter space.

Definition 3.3 (Dominated Family). A parametric family is dominated if there exists a σ -finite measure μ such that $P_\theta \ll \mu$ for all $\theta \in \Theta$.

Definition 3.4 (Likelihood Function). For a dominated family with densities $p_\theta = \frac{dP_\theta}{d\mu}$, the likelihood function is:

$$L : \mathcal{X} \times \Theta \rightarrow [0, \infty), \quad L(x, \theta) = p_\theta(x)$$

We interpret L two ways: fixed x gives a function of θ (the likelihood); fixed θ gives the density.

Definition 3.5 (Conditional Parametric Model). A collection of conditional probability measures:

$$\mathcal{P} = \{P_\theta(\cdot|x) : \theta \in \Theta, x \in \mathcal{X}\}$$

where each $P_\theta(\cdot|x)$ is a probability measure on $(\mathcal{Y}, \mathcal{C})$.

Definition 3.6 (Conditional Density). If $P_\theta(\cdot|x) \ll \nu$ for all x , then:

$$p_\theta(y|x) = \frac{dP_\theta(\cdot|x)}{d\nu}(y)$$

is the conditional density with respect to measure ν on $\mathcal{Y}$.

4 The Supervised Learning Model

Let (X, Y) be a pair of random variables. Supervised learning aims to model the conditional distribution $P_{Y|X}$. A common structural assumption is that the relationship can be written as $Y = f(X) + \epsilon$, where $f : \mathcal{X} \rightarrow \mathcal{Y}$ is a deterministic function and ϵ is a zero-mean noise variable independent of X .

Proposition 4.1 (The Function f as Conditional Expectation). The structural assumption implies that the function f is the conditional expectation of Y given X .

Proof. By linearity of expectation and independence of ϵ from X :

$$E[Y|X = x] = E[f(x) + \epsilon|X = x] = f(x) + E[\epsilon|X = x] = f(x).$$

Thus, learning the function f is equivalent to estimating the conditional mean of Y . $\square$

5 The I.I.D. Data Assumption

In supervised learning, we are given a dataset $\mathcal{D} = \{(x_1, y_1), \dots, (x_n, y_n)\}$. We assume these are **realizations** of n independent and identically distributed (i.i.d.) random variables, $(X_1, Y_1), \dots, (X_n, Y_n)$, each having the same joint probability distribution P_{XY} .

Definition 5.1 (I.I.D. Random Variables). The random pairs $(X_1, Y_1), \dots, (X_n, Y_n)$ are i.i.d. if their joint density function $p^{(n)} : (\mathcal{X} \times \mathcal{Y})^n \rightarrow [0, \infty)$ is the product of their identical marginal density functions:

$$((x_1, y_1), \dots, (x_n, y_n)) \mapsto \prod_{i=1}^n p_{XY}(x_i, y_i)$$

Proposition 5.1 (Consequences for Conditional Distributions). If the random variables (X_i, Y_i) are i.i.d., then conditioned on the realizations $X_1 = x_1, \dots, X_n = x_n$, the random variables $Y_1, \dots, Y_n$ are **conditionally independent**. However, their conditional distributions, $P_{Y_i|X_i=x_i}$, are **not** identically distributed unless all x_i are identical.

Proof. The joint density evaluated at the data point is $\prod_{i=1}^n p_{XY}(x_i, y_i)$. Applying the chain rule gives:

$$\prod_{i=1}^n p_{Y|X}(y_i|x_i) p_X(x_i) = \left(\prod_{i=1}^n p_{Y|X}(y_i|x_i) \right) \left(\prod_{i=1}^n p_X(x_i) \right)$$

The first term is the joint conditional density of $(Y_1, \dots, Y_n)$. Its product form proves conditional independence. However, the function $p_{Y|X}(\cdot|x_i) : y \mapsto p_{Y|X}(y|x_i)$ depends on the value x_i , so the conditional distributions are distinct for different x_i . $\square$

6 Conditional Maximum Likelihood Estimation

The Maximum Likelihood Estimation (MLE) principle prescribes choosing the parameter θ from a model family $\{p_\theta(\cdot|\cdot) : \theta \in \Theta\}$ that makes the observed data most probable.

Definition 6.1 (Conditional Likelihood). *Given data $(x_1, y_1), \dots, (x_n, y_n)$, the conditional likelihood is:*

$$L_n^{\text{cond}}(\theta) = \prod_{i=1}^n p_\theta(y_i|x_i)$$

Remark 6.1 (Classical vs. Conditional MLE). **Classical Statistics (Joint MLE):**

- Data: $x_1, \dots, x_n$
- Model: $X_i \stackrel{\text{i.i.d.}}{\sim} P_\theta$ (same distribution for all)
- Likelihood: $L(\theta) = \prod_{i=1}^n p_\theta(x_i)$

Machine Learning (Conditional MLE):

- Data: $(x_1, y_1), \dots, (x_n, y_n)$
- Model: $(X_i, Y_i) \stackrel{\text{i.i.d.}}{\sim} P_{X,Y}$ but we only model $P_{Y|X}$
- Conditional Likelihood: $L^{\text{cond}}(\theta) = \prod_{i=1}^n p_\theta(y_i|x_i)$
- Key: We use $p_\theta(y_i|x_i)$ (conditional density), NOT $p_\theta(x_i, y_i)$ (joint density)

The x_i are treated as fixed/given. We don't need to model how the features were generated.

Definition 6.2 (Conditional Log-Likelihood). *Given a dataset $\mathcal{D} = \{(x_1, y_1), \dots, (x_n, y_n)\}$, the conditional log-likelihood is the function $\ell_n : \Theta \rightarrow \mathbb{R}$ defined by:*

$$\ell_n(\theta) = \sum_{i=1}^n \log p_\theta(y_i|x_i)$$

The sum structure follows from conditional independence given in Proposition 5.1.

Remark 6.2 (ML Perspective: From Integrals to Sums). While measure theory defines densities via integrals ($P_X(A) = \int_A p(x) d\mu(x)$), machine learning works exclusively with finite and countable sums:

- The outer sum $\sum_{i=1}^n \log p_\theta(y_i|x_i)$ is finite over the training set
- For discrete outputs (classification), each $p_\theta(y|x) = \frac{\exp(f_\theta^{(y)}(x))}{\sum_{k=1}^K \exp(f_\theta^{(k)}(x))}$ is a ratio of finite sums
- For continuous outputs (regression), $p_\theta(y|x)$ is evaluated pointwise, not integrated

Yet we treat p_θ as a continuous, differentiable function of θ to enable automatic differentiation: gradients $\nabla_\theta \ell_n(\theta)$ are computed via backpropagation through the computational graph, not by differentiating integrals.

Definition 6.3 (Maximum Likelihood Estimator (MLE)). *The MLE, denoted $\hat{\theta}_n$, is:*

$$\hat{\theta}_n = \arg \max_{\theta \in \Theta} \ell_n(\theta) = \arg \max_{\theta \in \Theta} \sum_{i=1}^n \log p_\theta(y_i|x_i)$$

where $\arg \max_{\theta \in \Theta} f(\theta) = \{\theta^* \in \Theta : f(\theta^*) \geq f(\theta) \text{ for all } \theta \in \Theta\}$ denotes the set of maximizers.

Definition 6.4 (Negative Log-Likelihood Loss). *For a conditional density $p_\theta(y|x)$, define:*

$$\ell(\theta; x, y) = -\log p_\theta(y|x)$$

Proposition 6.1 (Equivalence of MLE and Risk Minimization). *Maximum conditional likelihood equals minimum empirical risk:*

$$\hat{\theta}_{MLE} = \arg \max_{\theta} \prod_{i=1}^n p_\theta(y_i|x_i) = \arg \min_{\theta} \sum_{i=1}^n \ell(\theta; x_i, y_i)$$

Remark 6.3 (The Origin of Loss Functions). The MLE principle directly yields loss functions from probabilistic assumptions:

- Maximizing $\ell_n(\theta)$ is equivalent to minimizing its negative
- The sum $\sum_{i=1}^n (-\log p_\theta(y_i|x_i))$ is the *empirical risk*
- The term $-\log p_\theta(y_i|x_i)$ is the *loss function* for one data point
- Each distributional assumption yields a specific loss function

Thus, the choice of loss function is not arbitrary—it encodes our beliefs about the conditional distribution of the data.

7 Parameterizing Distributions with Functions

We use a flexible function, parameterized by θ , to define the conditional density. This creates a chain of mappings:

1. **The Core Function (e.g., Neural Network):** A function $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^K, x \mapsto f_\theta(x)$, whose structure is defined by the parameters θ . A closed-form expression for the mapping is typically not available.
2. **The Parameter-Generating Function:** A function $g : \mathbb{R}^K \rightarrow \mathcal{P}_\mathcal{S}$, where $\mathcal{P}_\mathcal{S}$ is the space of parameters for a chosen distribution family (e.g., the mean for a Gaussian, or the probability vector for a Categorical). An example is the softmax function.
3. **The Conditional Density Function:** For a fixed x , the value $g(f_\theta(x))$ provides the parameters for the final conditional density function, $p_\theta(\cdot|x) : \mathcal{Y} \rightarrow [0, \infty)$.

8 Derivations from Probabilistic Models

Example 8.1 (Gaussian Conditional Yields Squared Loss). **Probabilistic Model:** Assume $Y|X = x \sim \mathcal{N}(\mu = f_\theta(x), \sigma^2)$ where $f_\theta : \mathcal{X} \rightarrow \mathbb{R}$ is a parametric function.

Conditional Density: With respect to Lebesgue measure on $\mathbb{R}$:

$$p_\theta(y|x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - f_\theta(x))^2}{2\sigma^2}\right)$$

Negative Log-Likelihood Loss:

$$\ell(\theta; x, y) = -\log p_\theta(y|x) = \frac{(y - f_\theta(x))^2}{2\sigma^2} + \frac{1}{2} \log(2\pi\sigma^2)$$

For fixed σ^2 , minimizing ℓ is equivalent to minimizing squared error:

$$\arg \min_{\theta} \sum_{i=1}^n \ell(\theta; x_i, y_i) = \arg \min_{\theta} \sum_{i=1}^n (y_i - f_\theta(x_i))^2$$

Intuition: The Gaussian assumption implies that errors $\epsilon_i = y_i - f_\theta(x_i)$ are i.i.d. normal with mean 0. Large errors receive quadratic penalty, making this loss sensitive to outliers. The MLE for linear $f_\theta(x) = \theta^T x$ recovers ordinary least squares regression.

Example 8.2 (Laplace Conditional Yields Absolute Loss). **Probabilistic Model:** The Laplace distribution with location μ and scale $b > 0$ has density:

$$p(y) = \frac{1}{2b} \exp\left(-\frac{|y - \mu|}{b}\right)$$

Assume $Y|X = x$ follows a Laplace distribution with location $f_\theta(x)$ and scale b :

$$p_\theta(y|x) = \frac{1}{2b} \exp\left(-\frac{|y - f_\theta(x)|}{b}\right)$$

Negative Log-Likelihood Loss:

$$\ell(\theta; x, y) = -\log p_\theta(y|x) = \frac{|y - f_\theta(x)|}{b} + \log(2b)$$

For fixed b , minimizing ℓ is equivalent to minimizing absolute error (L1 loss):

$$\arg \min_{\theta} \sum_{i=1}^n \ell(\theta; x_i, y_i) = \arg \min_{\theta} \sum_{i=1}^n |y_i - f_\theta(x_i)|$$

Intuition: The Laplace distribution has heavier tails than the Gaussian, making it more robust to outliers. This corresponds to the absolute loss being less sensitive to large errors than squared loss.

Example 8.3 (Bernoulli Conditional Yields Binary Cross-Entropy). **Probabilistic Model:** For binary $Y \in \{0, 1\}$, assume $Y|X = x \sim \text{Bernoulli}(p_\theta(x))$ where $p_\theta(x) = \sigma(f_\theta(x))$ and $\sigma : \mathbb{R} \rightarrow (0, 1)$ is the logistic (sigmoid) function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Conditional PMF: With respect to counting measure on $\{0, 1\}$:

$$p_\theta(y|x) = p_\theta(x)^y (1 - p_\theta(x))^{1-y}$$

Negative Log-Likelihood Loss:

$$\ell(\theta; x, y) = -\log p_\theta(y|x) = -y \log p_\theta(x) - (1 - y) \log(1 - p_\theta(x))$$

This is the **binary cross-entropy** loss. Substituting $p_\theta(x) = \sigma(f_\theta(x))$:

$$\ell(\theta; x, y) = -y \log(\sigma(f_\theta(x))) - (1 - y) \log(1 - \sigma(f_\theta(x)))$$

Example 8.4 (Categorical Conditional Yields Softmax Loss). **Probabilistic Model:** For $Y \in \{1, \dots, K\}$, assume $Y|X = x$ follows a categorical distribution parametrized via the softmax function:

$$P_\theta(Y = k|x) = \frac{\exp(f_\theta^{(k)}(x))}{\sum_{j=1}^K \exp(f_\theta^{(j)}(x))}$$

where $f_\theta^{(k)} : \mathcal{X} \rightarrow \mathbb{R}$ for $k = 1, \dots, K$ are parametric functions (often called logits).

Conditional PMF: With respect to counting measure on $\{1, \dots, K\}$, the probability of observing class y is:

$$p_\theta(y|x) = \frac{\exp(f_\theta^{(y)}(x))}{\sum_{k=1}^K \exp(f_\theta^{(k)}(x))}$$

Negative Log-Likelihood Loss:

$$\ell(\theta; x, y) = -\log p_\theta(y|x) = -f_\theta^{(y)}(x) + \log \sum_{k=1}^K \exp(f_\theta^{(k)}(x))$$

This is the **categorical cross-entropy** (or softmax) loss.

One-Hot Encoding: If Y is represented as a one-hot vector $\mathbf{y} = (y_1, \dots, y_K)$ with $y_k = \mathbf{1}_{Y=k}$, and we write $p_k(x) = P_\theta(Y = k|x)$, then:

$$\ell(\theta; x, \mathbf{y}) = -\sum_{k=1}^K y_k \log p_k(x)$$

Conclusion

The principle of maximum likelihood provides a unified theoretical foundation for the objective functions used in supervised learning. The key insights are:

1. **Every loss function encodes distributional assumptions:** The choice of loss is not arbitrary—it corresponds to a specific assumption about the conditional distribution $P_{Y|X}$.
 - Squared loss $\leftrightarrow$ Gaussian conditional distribution
 - Absolute loss $\leftrightarrow$ Laplace conditional distribution
 - Cross-entropy $\leftrightarrow$ Bernoulli/Categorical conditional distribution
2. **Conditional vs. joint modeling:** Machine learning focuses on modeling $P_{Y|X}$ (conditional MLE) rather than $P_{X,Y}$ (joint MLE). This allows us to:
 - Train on diverse features without modeling their distribution
 - Focus computational effort on the prediction task
 - Make predictions for new feature values
3. **Optimization as statistical estimation:** Minimizing empirical risk via gradient descent is fundamentally solving a statistical estimation problem. The MLE principle transforms probabilistic modeling into concrete optimization objectives.

This perspective reveals that training a machine learning model is a well-defined statistical procedure to find the parameters of a conditional probability model that best explain the observed data.

Glossary

Mathematical Terminology	Common Machine Learning Terminology
Parameter Estimation	Training a model
Parameters (θ)	Weights and Biases
The function $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^K$	The Neural Network / Model
The value $f_\theta(x)$	The model's output / prediction / logits
Objective Function	Loss Function / Cost Function (on a dataset)
Loss Function (e.g., $-\log p_\theta(y x)$)	Loss Function (for a single example)
The function $p_\theta(\cdot x) : \mathcal{Y} \rightarrow [0, \infty)$	The model's predicted distribution
The value $p_\theta(y x)$	The model's predicted probability/density for y